WNDRVR
SUPPORT NETWORK

VXWORKS 7 KERNEL APPLICATION TUTORIAL - COMMAND LINE, SR0650

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VxWorks 7 Kernel Application Tutorial - Command Line, SR0650

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1. KERNEL APPLICATION TUTORIAL FROM THE COMMAND LINE

This VxWorks® 7 Kernel Application Tutorial - Command Line, provides instructions on how to create, build, and run a Hello World kernel application from the command line.

VxWorks applications that execute in the kernel are created as relocatable object modules and are executed as a set of tasks running in kernel space. They can be referred to most specifically as kernel-based application modules, but it is often convenient to refer to them simply as kernel application modules or kernel applications. They are sometimes referred to as downloadable kernel modules (DKMs) even when they are linked with the kernel. A kernel application is derived from a DKM project that includes user source code.

From the command line, you must base your DKM project on a VxWorks source build (VSB) project. The VSB produces binary kernel libraries that are used by VIP and DKM projects.

For this tutorial, the workflow is as follows:

1. Create and build a VSB.
2. Create a VIP based on the VSB and build the VIP.
3. Create a DKM project based on the VSB.
4. Add code to the DKM project and build the project to obtain a kernel application.
5. Connect to the VxWorks simulator and boot VxWorks.
6. Load and run the kernel application.
7. Observe the results.

Step 5 may be achieved using one of two approaches. Both approaches are documented here. Step 6 also can be achieved using one of two documented approaches.

For more information, see Environment Variables for the VxWorks Development Tools.

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2. CONFIGURING AND BUILDING VXWORKS SOURCE FROM THE COMMAND LINE

Before beginning any VxWorks development, generate the VxWorks kernel libraries by creating, configuring, and building a VxWorks source build (VSB) project.

About This Task

For this tutorial, create a VSB based on a board support package (BSP) for the VxWorks simulator.

Procedure

1. Start the wrtool utility.
   1. Open a command window and access the *installDir* **/workbench-4** directory.

      To view the startup command options for the wrtool utility, type **wrtool /?**.

      2. Type wrtool at the prompt to invoke the wrtool utility.

      The active workspace is set by the **WIND_WRTOOL_WORKSPACE** environment variable.

      ```
      C:/WindRiver/workbench-4> wrtool
      Active workspace is 'C:\WindRiver\workbench-4\workspace'.
      WrTool, version 4.12.0
      Copyright (c) 2013 - 2016 Wind River Systems, Inc.
      workbench-4>
      ```

      You can also specify the active workspace with the -data command option.

      ```
      C:/WindRiver/workbench-4> wrtool -data C:\WindRiver\workbench-4\workspace
      Active workspace is 'C:\WindRiver\workbench-4\workspace'.
      WrTool, version 4.12.0
      Copyright (c) 2013 - 2016 Wind River Systems, Inc.
      workbench-4>
      ```

      To view the list of command groups for the wrtool utility, type help.

      To view the list of commands for the wrtool session, type help builtin.

2. Access the active workspace directory.
3. Create a VSB project for the VxWorks simulator using the appropriate BSP for your host type.

Host OS	BSP
Windows	**vxsim_windows**_*x_x_x_x*
Linux	**vxsim_linux**_*x_x_x_x*

For this example, use the following syntax:

```
prj vsb create -bsp bspName -S name
```

To get help for the vsb create command option, type help prj vsb create.

To view the list of available BSPs, type prj vsb listBsps.

The following command creates a VSB for the Windows VxWorks simulator BSP version 1.0.0.0 and uses the -S option to accept all the default settings and not proceed with the interactive configuration process:

```
workspace> prj vsb create -bsp vxsim_windows_1_0_2_11 -S VxSimVSBwrtool2

Defaulting VSB to ILP32 to match BSP selected
Creating VSB Project ...

starting VSB configuration...

#
# configuration written to vsb.config
#

Checking layers required by BSP vxsim_windows_1_0_2_11.
Nothing to add all requested layers enabled

done

VSB Project created in : C:/WindRiver/workbench-4/workspace/VxSimVSBwrtool2

This VSB supports the following BSPs: vxsim_windows_1_0_2_11


workspace>
```

For this tutorial, accept the default configuration options and proceed directly to building the project.

Note: To view and configure the project, access the VSB project directory and use the following commands:

- To view the enabled project options, type prj vsb list.
- To view the available project options, type prj vsb listAll.
- To get help on configuring the project options, type help prj vsb config.

4. Build the VSB project using the prj build command and observe the build results.

```
workspace> prj build VxSimVSBwrtool
```

To get help for the build command option, type help prj build.

The project resides in the workspace folder. For example:

installDir **/workbench-4/workspace/VxSimVSBwrtool**

3. CONFIGURING AND BUILDING A VXWORKS KERNEL IMAGE FROM THE COMMAND LINE

To create a VxWorks kernel image, build VxWorks from binary components by creating a VxWorks image project (VIP) based on a VxWorks source build (VSB) project.

About This Task

For this tutorial, base the project on the **VxSimVSBwrtool** project.

Procedure

1. In the workspace directory, create a VIP project using the prj vip create command.

   For this example, use the following syntax:

   ```
   prj vip create -vsb vsbDir bspName toolName projDir
   ```

   Where *bspName* is the BSP for the VSB project and *toolName* is the toolchain.

   To get help for the vip create command option, type help prj vip create.

   The following command creates a VIP for the Windows VxWorks simulator named **VxSimVIPwrtool** that is based on the **VxSimVSBwrtool** project and uses the **gnu** tool chain:

   ```
   workspace> prj vip create -vsb VxSimVSBwrtool vxsim_windows_1_0_0_0 gnu VxSimVIPwrtool
   ...
   Project created in : C:/WindRiver/workspace/VxSimVIPwrtool/VxSimVIPwrtool.wpj
   ```

   For this tutorial, accept the default component and parameter configuration, and build the project.

   Note: To view and configure the project components, access the VIP project directory and use the following commands:

   - To view the included components, type prj vip component list.
   - To view the available components, type prj vip component list all.
   - To change the component configuration, use the prj vip component add and prj vip component remove commands.
   - To get help on configuring the VIP component parameter values, use the help prj vip parameter command.

2. To build the project, access the VIP directory and use the prj build command. For example:

   ```
   workspace> cd VxSimVIPwrtool
   VxSimVIPwrtool> prj build
   ```

   After the build completes, the image resides in the workspace folder. For example:

   installDir **/workbench-4/workspace/VxSimVIPwrtool/default/vxWorks**

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4. CREATING A DOWNLOADABLE KERNEL MODULE PROJECT FROM THE COMMAND LINE

From the command line, a kernel application must be based on a VxWorks source build (VSB) project.

About This Task

For this tutorial, base the DKM project on the **VxSimVSBwrtool** project.

Procedure

1. Access the workspace directory and use the prj list command to list existing VSB and VIP projects. For example:

```
VxSimVIPwrtool> cd ..
workspace> prj list
VxSimVIPwrtool (Wind River VxWorks 7 Image Project)
VxSimVSBwrtool (Wind River VxWorks 7 Source Build Project)
```

2. Create a DKM project using the following syntax:

```
workspace> prj dkm create -vsb vsbDir dkmProj
```

   To get help for the dkm create command option, type help prj dkm create.

   The following command creates a DKM project named **myDKMwrtool** based on the VSB project **VxSimVSBwrtool**:

```
workspace> prj dkm create -vsb VxSimVSBwrtool myDKMwrtool
```

   The project is created in the workspace folder. For example:

   installDir **/workbench-4/workspace/myDKMwrtool**

   A stub C source file named **dkm.c** is automatically added to the DKM project directory.

5. ADDING SOURCE CODE TO THE DOWNLOADABLE KERNEL MODULE PROJECT

For this tutorial, you can simply add source code to the stub C source file and build the project to create a kernel application.

About This Task

Add code to the **dkm.c** file to create a kernel application that displays `Hello World! From the kernel...` on the console.

Procedure

1. Access the DKM project directory.
2. Using a text editor, add a **stdio.h** include directive and a call to printf( ) to the **dkm.c** file.

   The source code should be as follows:

```
#include "vxWorks.h"
#include "stdio.h"

void start(void) {
  printf("Hello World!\nFrom the kernel...\n");
}
```

3. Save the file.
4. Build the project and observe the successful completion of the build.

```
myDKMwrtool> prj build
```

5. Exit the wrtool utility:

```
myDKMwrtool> exit
```

6. WHERE TO NEXT?

After creating and adding code to a VxWorks application using command line, based on a tutorial, see the following steps to connect to your target, load your DKM and call C functions in the kernel.

1. To connect to a remote file system see *VxWorks 7 File Systems and I/O Programmer's Guide: Accessing Remote File Systems from VxWorks*.
2. To load a DKM object see *VxWorks 7 Kernel Shell User's Guide:Loading C Modules into the Kernel*.
3. To call C functions see *VxWorks 7 Kernel Shell User's Guide: Calling C Functions From the C Interpreter*.

For more information on making a target connection and debugging with the WRDBG (Wind River DeBuG) shell see the second Debug Session Example in *WRDBG Debug Shell Reference: Getting Started*

For more information on VxWorks projects, see the *VxWorks 7 Configuration and Build Guide* and the *Kernel Application and Real-Time Process Application Projects Guide*.

For more information on getting started with VxWorks 7, see the getting started documentation from the Wind River Support Network at Getting Started.